Title	Loan Growth Model @ Loan Projection
Department	Strategic Planning

No.	Name	Date	Activity
1.	Muhammad Syahid Bin Abd Halid	22/8/2023	Create Documentation
2.			Update

EXECUTIVE SUMMARY

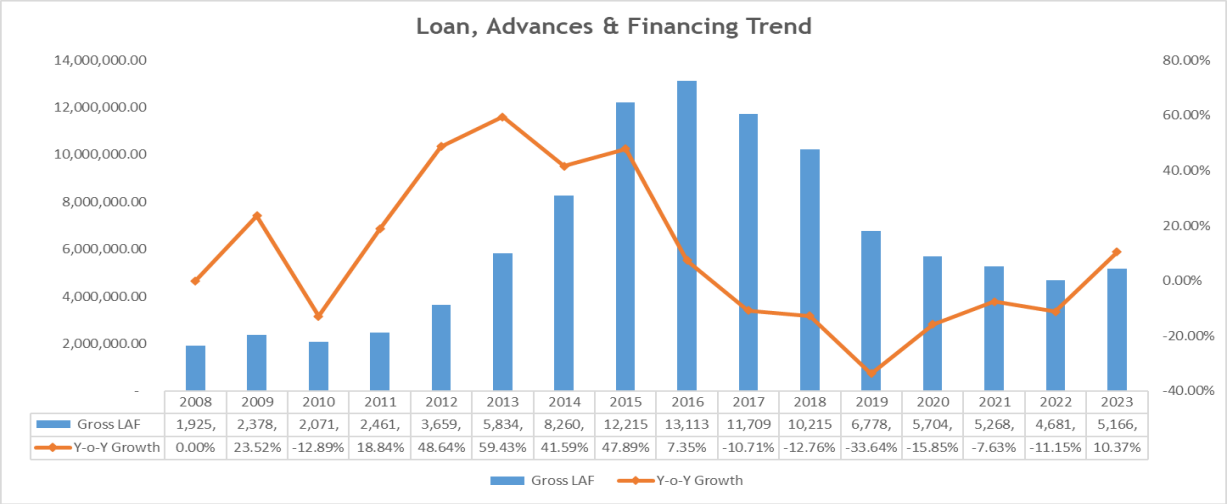
1. Background

Table below shows EXIM Bank's Loan, Advances & Financing (LAF, RM'000) and Y-o-Y Growth throughout 2008 until 2023:

Year	LAF (RM'000)	Y-o-Y LAF
2008	1,925,325.00	0.00%
2009	2,378,090.00	23.52%
2010	2,071,481.00	-12.89%
2011	2,461,820.00	18.84%
2012	3,659,240.00	48.64%
2013	5,834,086.00	59.43%
2014	8,260,354.00	41.59%
2015	12,215,867.00	47.89%

Year	LAF (RM'000)	Y-o-Y LAF
2016	13,113,129.00	7.35%
2017	11,709,044.00	-10.71%
2018	10,215,188.00	-12.76%
2019	6,778,920.00	-33.64%
2020	5,704,223.00	-15.85%
2021	5,268,867.00	-7.63%
2022	4,681,502.00	-11.15%
2023	5,166,986.35	10.37%

Figure below shows EXIM Bank's Loan, Advances & Financing (LAF, RM'000) and Y-o-Y Growth throughout 2008 until 2023:



Using Statistical and Mathematical method we want to obtain as per below:

1- What Macroeconomics variable impact most in the loan growth.
2- What is the benchmark of the growth.
3- What is expected until 2026.

2. Detailed

After carefully examining the trend of LAF in the figure above, it shows the growth of LAF is non-linear thus the most accurate best fit line to project the LAF growth would be a non-2-D model assuming the macroeconomics parameter are the variable for the LAF growth model.

1) Macroeconomics Parameter

- USD Fed (Actual)
- FX (1 USD)
- OPR (Bank Negara)
- GDP (Constant LCU)
- KLSE Index @ FTSE Bursa MY KLCI Index

2) Coefficient Correlation

	LAF	USD Fed	FX	OPR	GDP	KLSE Index
LAF	1.00					
USD Fed	0.20	1.00				
FX	0.69	0.56	1.00			
OPR	0.34	0.17	-0.10	1.00		
GDP	0.52	0.67	0.82	-0.15	1.00	
KLSE Index	0.58	0.07	0.17	0.12	0.43	1.00

Interpretation: All the parameters are positively correlated with the LAF growth and FX shows the most correlation with 0.69 bps.

3) Granger Causality Test

H0: The Macroeconomics Parameter, does NOT Granger cause LAF. Assuming a significance level of 0.05, if the p-value is lesser than 0.05 then we do NOT reject the null hypothesis that X does NOT granger cause Y.

- USD Fed (Actual)

	LAF, X	USD Fed, X
LAF, Y	1.0	0.0005
USD Fed, Y	0.0	1.0000

Interpretation: When X = USD Fed and Y = LAF is 0.0005, we reject the null hypothesis and conclude that X granger caused Y.

- FX (1 USD)

	LAF, X	FX, X
LAF, Y	1.0000	0.0017
FX, Y	0.2165	1.0000

Interpretation: When X = FX and Y = LAF is 0.0017, we reject the null hypothesis and conclude that X granger caused Y.

- OPR (Bank Negara)

	LAF, X	OPR, X
LAF, Y	1.000	0.1824
OPR, Y	0.005	1.000

Interpretation: When X = OPR and Y = LAF is 0.1823, we accept the null hypothesis and conclude that X does not granger cause Y.

- GDP (Constant LCU)

	LAF, X	GDP, X
LAF, Y	1.0000	0.0791
GDP, Y	0.0014	0.9998

Interpretation: When X = GDP and Y = LAF is 0.0791, we accept the null hypothesis and conclude that X does not granger cause Y.

- KLSE Index @ FTSE Bursa MY KLCI Index

	LAF, X	KLSE Index, X
LAF, Y	1.0000	0.001
KLSE Index, Y	0.0063	1.000

Interpretation: When X = KLSE Index and Y = LAF is 0.001, we reject the null hypothesis and conclude that X granger caused Y.

In conclusion, USD Fed, FX and KLSE Index will likely predict LAF Growth.

4) Multiple Linear Regression

Formula:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_i X_i$$

Y : Dependent variable
β_0 : Intercept
β_i : Slope for X_i
X = Independent variable

Model 1: USD Fed, FX and KLSE Index
Model 2: All Macroeconomics

- Regression Statistics

Model	1	2
Multiple R	0.86	0.97
R Square	0.74	0.95
Adjusted R Square	0.67	0.92
Standard Error	2,132,646.04	1,046,543.51
Observations	16.00	16.00

Interpretation: Though both Multiple R show a high predictive power against the macroeconomics parameter, Model 1 is slightly lower with 0.86 bps while Model 2 with 0.97 bps.

- ANOVA

Model 1	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	3	155,114,223,545,299.00	51,704,741,181,766.30	11.37	0.0008
Residual	12	54,578,149,476,639.60	4,548,179,123,053.30		
Total	15	209,692,373,021,938.00			

Model 2	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	5	198,739,839,876,406.00	39,747,967,975,281.20	36.29	0.00
Residual	10	10,952,533,145,532.40	1,095,253,314,553.24		
Total	15	209,692,373,021,938.00			

Interpretation: Significance F indicates low chance of random occurrence or false to the model with the macroeconomics parameter for both model by value less than 0.15 bps.

- Confidence Level

Model 1	*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-26,754,030.97	5,697,817.81	-4.70	0.00	- 39,168,509.51	-14,339,552.43
USD Fed	-1,002,958.70	445,479.89	-2.25	0.04	-1,973,575.99	-32,341.40
FX	5,813,406.98	1,369,918.84	4.24	0.00	2,828,610.23	8,798,203.73
KLSE Index	7,674.60	2,391.53	3.21	0.01	2,463.92	12,885.29

Model 2	*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-37,877,784.97	3,363,713.83	-11.26	0.00	-45,372,606.45	-30,382,963.48
USD Fed	-841,103.15	343,033.89	-2.45	0.03	-1,605,430.29	-76,776.00
FX	8,742,813.50	1,076,764.24	8.12	0.00	6,343,633.27	11,141,993.72
OPR	2,594,894.85	635,179.14	4.09	0.00	1,179,627.54	4,010,162.16
GDP	-0.00	0.00	-2.30	0.04	-0.00	-0.00
KLSE Index	9,103.88	1,620.49	5.62	0.00	5,493.21	12,714.56

Interpretation: P-value indicates all the parameters have a high predictive power with the macroeconomics parameter by value less than 0.15 bps.

Equation Model 1:

$$y = -26{,}754{,}030.97 - 1{,}002{,}958.70x_1 + 5{,}813{,}406.98x_2 + 7{,}674.60x_3$$

Equation Model 2:

$$y = -37{,}877{,}784.97 - 841{,}103.15x_1 + 8{,}742{,}813.50x_2 + 2{,}594{,}894.85x_3 + 9{,}103.88x_5$$

- Error Rate Model 1

Year	LAF Actual (RM'000)	LAF Projection (RM'000)	A. Growth	P. Growth	Error Rate
2008	1,925,325.00	- 138,419.35	0.00%	0.00%	-1490.94%
2009	2,378,090.00	2,671,324.71	23.52%	-2029.88%	-10.98%
2010	2,071,481.00	2,577,903.34	-12.89%	-3.50%	-19.64%
2011	2,461,820.00	3,212,170.72	18.84%	24.60%	-23.36%
2012	3,659,240.00	3,736,395.24	48.64%	16.32%	-2.06%
2013	5,834,086.00	6,400,104.03	59.43%	71.29%	-8.84%
2014	8,260,354.00	6,829,983.97	41.59%	6.72%	20.94%
2015	12,215,867.00	10,684,977.34	47.89%	56.44%	14.33%
2016	13,113,129.00	11,172,322.21	7.35%	4.56%	17.37%
2017	11,709,044.00	9,145,396.29	-10.71%	-18.14%	28.03%
2018	10,215,188.00	7,771,889.99	-12.76%	-15.02%	31.44%
2019	6,778,920.00	7,475,264.06	-33.64%	-3.82%	-9.32%
2020	5,704,223.00	8,812,624.80	-15.85%	17.89%	-35.27%
2021	5,268,867.00	9,302,189.74	-7.63%	5.56%	-43.36%
2022	4,681,502.00	5,863,823.49	-11.15%	-36.96%	-20.16%
2023	5,166,986.35	5,926,171.78	10.37%	1.06%	-12.81%

Max	13,113,129.00	11,172,322.21	70.59%	71.29%	31.44%
Min	1,925,325.00	-138,419.35	-33.64%	-2029.88%	-1490.94%
Average	6,631,299.93	6,777,474.09	12.60%	-101.60%	-88.35%

- Error Rate Model 2

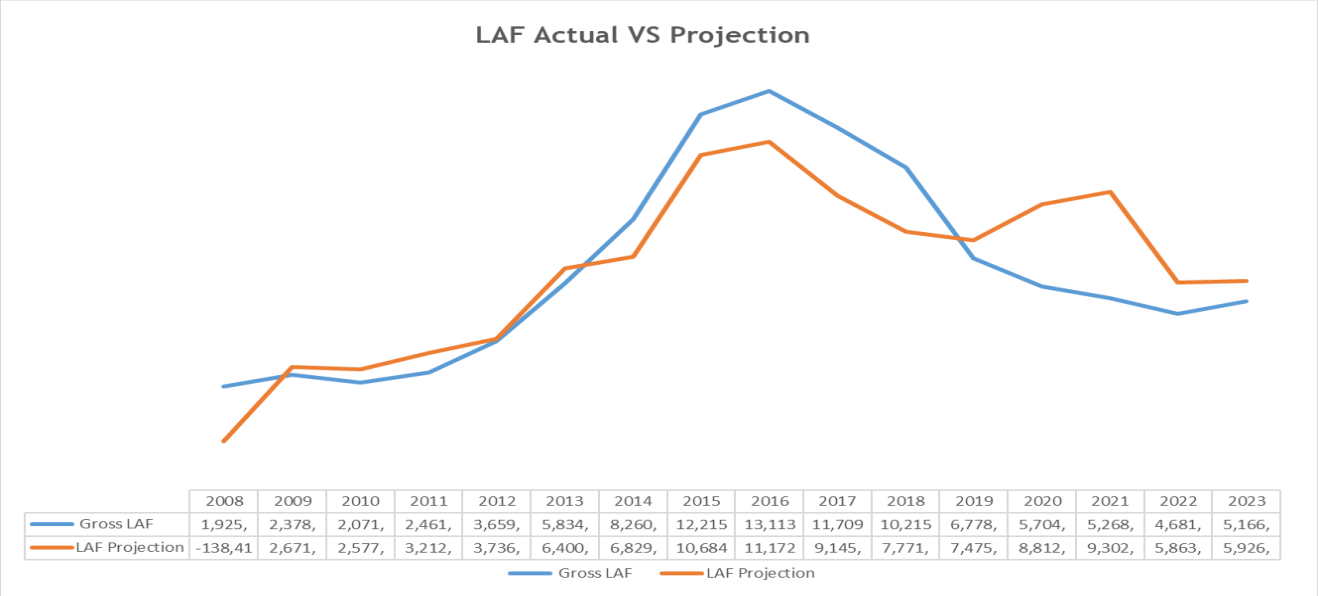
Year	LAF Actual (RM'000)	LAF Projection (RM'000)	A. Growth	P. Growth	Error Rate
2008	1,925,325.00	1,553,310.21	0.00%	0.00%	23.95%
2009	2,378,090.00	1,676,601.41	23.52%	7.94%	41.84%
2010	2,071,481.00	2,366,026.84	-12.89%	41.12%	-12.45%
2011	2,461,820.00	3,544,445.20	18.84%	49.81%	-30.54%
2012	3,659,240.00	3,516,674.56	48.64%	-0.78%	4.05%
2013	5,834,086.00	6,699,350.89	59.43%	90.50%	-12.92%
2014	8,260,354.00	7,730,705.78	41.59%	15.39%	6.85%
2015	12,215,867.00	13,393,976.24	47.89%	73.26%	-8.80%
2016	13,113,129.00	13,338,330.31	7.35%	-0.42%	-1.69%
2017	11,709,044.00	9,825,055.77	-10.71%	-26.34%	19.18%
2018	10,215,188.00	8,816,599.82	-12.76%	-10.26%	15.86%
2019	6,778,920.00	6,974,891.60	-33.64%	-20.89%	-2.81%

2020	5,704,223.00	5,295,664.81	-15.85%	-24.08%	7.71%
2021	5,268,867.00	5,835,548.45	-7.63%	10.19%	-9.71%
2022	4,681,502.00	5,280,152.29	-11.15%	-9.52%	-11.34%
2023	5,166,986.35	5,596,788.18	10.37%	6.00%	-7.68%

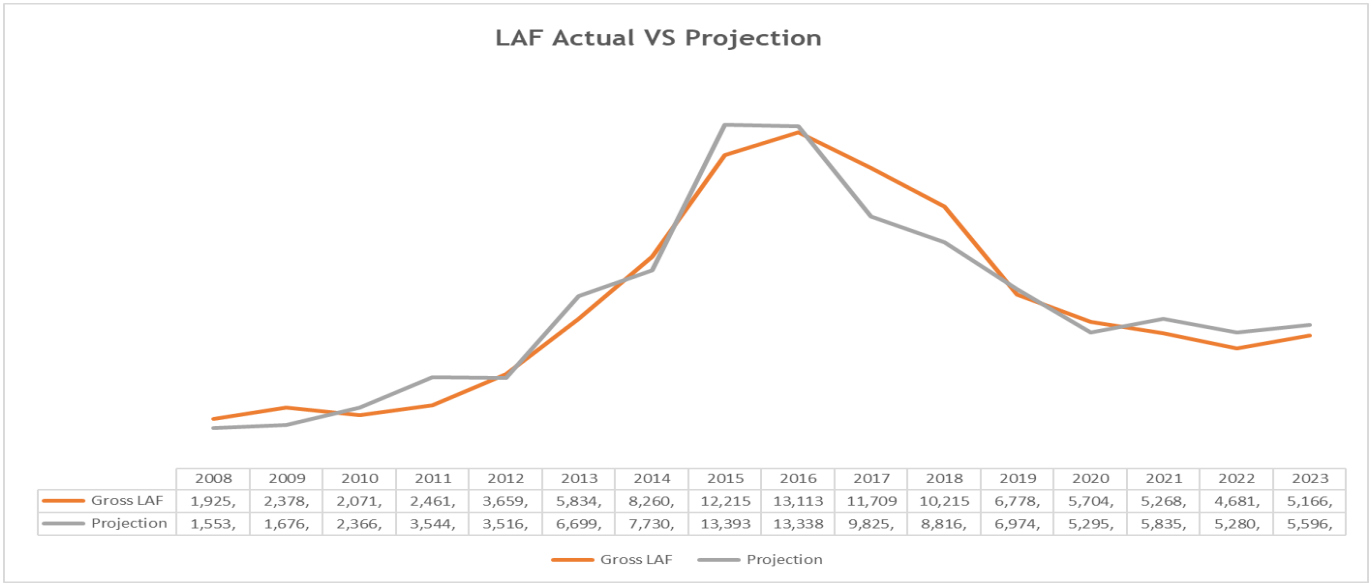
Max	13,113,129.00	13,393,976.24	70.59%	90.50%	41.84%
Min	1,925,325.00	1,553,310.21	-33.64%	-26.34%	-30.54%
Average	6,631,299.93	6,637,971.19	12.60%	14.65%	1.12%

Using the Regression Model, figure below shows the LAF Actual against the projection from 2008 until 2023 (RM'000) for both models:

Model 1:



Model 2:



3. Findings

Projection	Model 1	Model 2
Standard Deviation	2,132,646.04	1,046,543.51
Observation	16.00	16.00
MSE	533,161.51	261,635.88
Mean	6,777,474.09	6,637,971.19
Multiple R	0.86	0.97

From the table above assumption as per below:

- Multiple R shows Model 2 has high predictive power with the macroeconomics parameter with 0.97 bps compared to Model 1.

- MSE Model 2 shows a more accurate reflection towards the actual mean the with less spread out compared to Model 1.

4. Conclusion

From the findings above, we conclude to use Model 2 due to higher predictive power as our Lower Limit while Model 1 as our Upper Limit.

1- Though the most macroeconomics variable that gave impact to the loan growth is USD Fed, FX and KLSE Index, OPR and GDP also will likely predict LAF Growth as well.

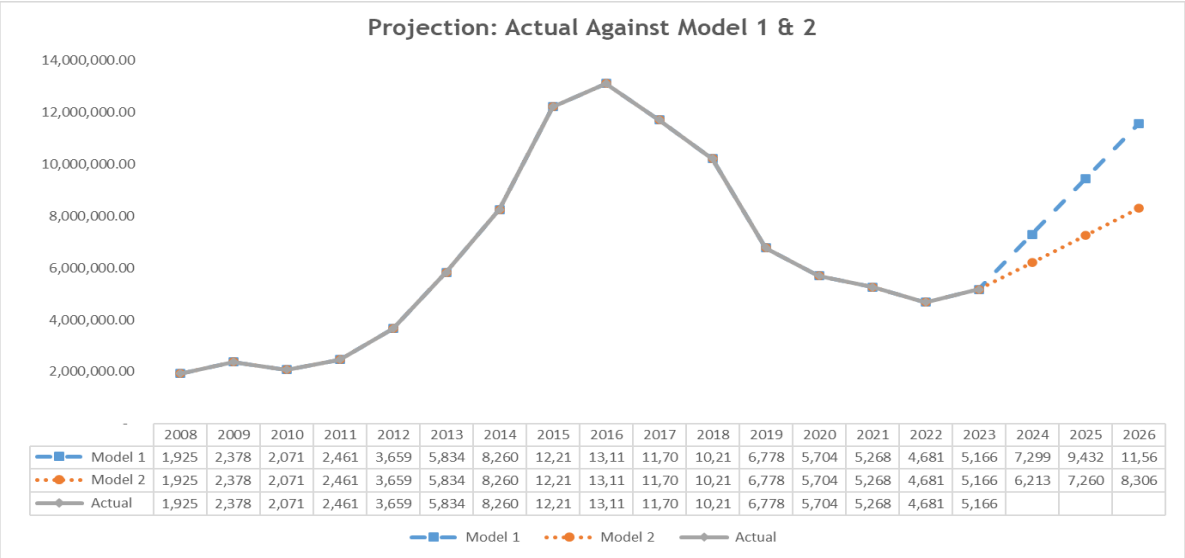
2- The benchmark of the targeted growth:

Model	Amount ('000)
1	+RM2,132,646.04
2	+RM1,046,543.51

3- What is expected growth until 2026.

Year	Model 1	Model 2	Growth (M.1)	Growth (M.2)
2023	5,166,986.35	5,166,986.35	10.37%	10.37%
2024	7,299,632.39	6,213,529.86	41.27%	20.25%
2025	9,432,278.43	7,260,073.37	29.22%	16.84%
2026	11,564,924.47	8,306,616.88	22.61%	14.42%

Using the Regression Model, figure below shows the LAF projection for Model 1 and Model 2 until 2026 (RM'000):



	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Model 1	1,925	2,378	2,071	2,461	3,659	5,834	8,260	12,21	13,11	11,70	10,21	6,778	5,704	5,268	4,681	5,166	7,299	9,432	11,56
Model 2	1,925	2,378	2,071	2,461	3,659	5,834	8,260	12,21	13,11	11,70	10,21	6,778	5,704	5,268	4,681	5,166	6,213	7,260	8,306
Actual	1,925	2,378	2,071	2,461	3,659	5,834	8,260	12,21	13,11	11,70	10,21	6,778	5,704	5,268	4,681	5,166			